
Observability Debt in Agent Evaluation: Toward a Telemetry Layer for AgenticOS

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Abstract

Agentic AI systems are evaluated today through three disjoint public infrastructures—leaderboards, trace corpora, and telemetry logs—none of the public sources we examine jointly exposes the fields a resource-aware operating layer would need: task identity, model identity, scaffold identity, outcome, token/cache usage, retries, latency, and billing. We treat this fragmentation as an empirical object and measure its consequences across three studies on public agent-evaluation data. First, using the Holistic Agent Leaderboard (HAL), raw dollar-cost rankings transfer across some benchmark pairs (Spearman $\rho=0.74\text{--}0.93$) but are largely explained by a stable displayed-label cost propensity (pooled model $R^2=82.0\%$); success-adjusted cost is markedly less portable (mean $\rho=0.59$ vs. 0.84). Second, using 977 exact task–model–matched trajectory pairs from Open-SWE-Traces, scaffold choice changes observable execution burden at fixed task and model (OpenHands/SWE-agent turn ratio 0.85 , $[0.82, 0.87]$), and the widely assumed “expensive failure” premium is not detected under exact task matching (80%-power threshold: premium fractions ≥ 0.65 ; observed $0.47\text{--}0.53$). Third, auditing two independent telemetry corpora (8,058 and 1,017 sessions), we characterize the resource anatomy—93.7% mean cache-read share, $4.29\times$ conditional tool-error cascades replicated across sources, top-1%-of-sessions holding 48.3% of input-token volume—that a scheduler or router would need but that no benchmark currently attaches to an outcome. These studies are complementary rather than linked: Studies 1–2 show that transferred raw-cost rankings and unconditioned failure premiums confound *which workloads were attempted* with *how agents behaved on them*, while Study 3 shows that the execution variables hidden beneath those aggregates are themselves heterogeneous, heavy-tailed, and state-dependent. From this evidence we derive a joint telemetry schema spanning identity, execution, resources, control, and outcome, and argue it belongs at the operating-system layer: the layer that controls execution and allocates resources must also observe what those allocations require.

1 Why This Belongs at the OS Layer

Agentic AI systems persist state, invoke tools, retry failed actions, and accumulate context across long horizons. Reasoning about their resource behavior—for scheduling, routing, cost budgeting, or governance—requires the same unified telemetry that operating systems have long provided for conventional workloads: a common schema linking *what ran*, *on what*, *under which policy*, *with what outcome*, and *at what resource cost*. We use *observability debt* to name the accumulated inability of public agent-evaluation infrastructure to jointly attribute resource consumption to task, model, execution policy, outcome, and system-level events. In every public source we examined, this debt prevents reliable attribution of observed resource differences to workload, model, scaffold, configuration, or execution behavior.

What makes this an *operating-system* responsibility rather than better benchmark bookkeeping? An agentic OS occupies a specific functional position. It observes *identity* (which task, model, scaffold, and policy configuration is running), *execution* (which tools act, what context accumulates), *resources* (tokens, cache, time, memory, money), *control* (retry, terminate, route, budget), and *outcome* (did the task succeed). The principle that places resource accounting inside conventional OSES applies verbatim: **the layer that controls execution and allocates resources must also observe what those allocation decisions require**. Without per-round token

41 components, a memory manager cannot budget context growth; without configuration identity, a router cannot
 42 learn *why* one label is cheap; without error-status streams, a governor cannot halt a spiraling session. Today,
 43 each of these decisions is made implicitly from aggregate dollar figures.

44 We treat three public data sources as stand-ins for what this observability layer would unify: a cost-and-outcome
 45 leaderboard (HAL), a task/model/scaffold-matched trajectory corpus (Open-SWE-Traces), and raw session-
 46 level usage telemetry (two independent corpora). Each exposes a different partial view. HAL links dollar cost
 47 to outcome but encrypts execution traces. Open-SWE-Traces exposes exact task/model/scaffold matching and
 48 resolved status but no tokens or billing. The telemetry corpora expose token, cache, and tool-call dynamics but
 49 no outcome or billing. **No public source we examined jointly exposes task, model, scaffold/configuration,**
 50 **outcome, token/cache usage, retries, latency, and billing,** the joint schema a resource-aware OS layer needs
 51 (Table 2).

52 Our three studies are therefore complementary rather than linked—each probes a different dimension of the
 53 gap, and none can be joined to another at the record level. (1) A coverage-aware cost-portability study (HAL)
 54 shows that aggregate dollar cost conflates displayed-label propensity with benchmark difficulty and decorrelates
 55 under success adjustment. (2) A matched-trajectory study shows that execution structure differs measurably
 56 across scaffolds, while a popular failure-cost heuristic turns out to measure workload composition. (3) A cross-
 57 source telemetry audit shows that the execution variables hidden beneath aggregate cost are themselves highly
 58 heterogeneous, heavy-tailed, and state-dependent. Together they triangulate one gap; no pair of them jointly
 59 identifies a causal mechanism.

60 2 Related Work

61 Kapoor et al. (2026) introduce HAL as cost-aware evaluation infrastructure spanning coding, web, science, and
 62 customer-service tasks (21,730 rollouts, roughly \$40,000 in evaluation cost). Kapoor et al. (2024) motivate joint
 63 accuracy–cost reporting and convex-hull frontier analysis, which we adopt as one of three decision-frontier def-
 64 initions. Ndzomga (2026) asks whether task subsets preserve agent rankings within a single benchmark; we in-
 65 stead ask whether an *observed cost signal* predicts a *different* workload. Benchmark² studies cross-benchmark
 66 rank consistency for static LLM benchmarks; our setting adds interactive scaffolds, dollar cost, success adjust-
 67 ment, and a held-out workload test. SWE-Effi reports that failed attempts consume substantially more resources
 68 than successful ones; our matched design tests that claim at fixed task and does not detect the premium once
 69 task-difficulty confounding is removed (Section 4). Open-SWE-Traces supplies the exact matching that makes
 70 this test possible but lacks token/billing fields, motivating Section 6.

71 3 Study 1: Does a Cost Signal Travel Across Workloads?

72 3.1 Data and cohort

73 We use the public `all_leaderboards_costs_HAL.csv` snapshot distributed with Ndzomga (2026): 242 dis-
 74 played evaluations across nine benchmarks, 11 scaffolds, and 29 displayed model labels. The CSV supplies
 75 accuracy and aggregate dollar cost but no token components, provider aliases, tool-call counts, retries, or per-
 76 task trajectories—the exact gap this paper is about. Most evaluations are single-run, so rollout-level uncertainty
 77 is unavailable.

78 The only scaffold repeated broadly enough is the displayed HAL Generalist Agent. Fixing it and matching exact
 79 label strings yields six workloads with adequate overlap—CORE-Bench Hard, GAIA, SciCode, ScienceAgent-
 80 Bench, SWE-bench Verified Mini, TAU-bench Airline—for a cohort of 86 rows and 25 labels (USACO has one
 81 Generalist observation and is excluded).

82 3.2 Raw cost transfers; the mechanism behind it does not

83 Across six pairs with ≥ 12 shared labels, raw dollar-cost Spearman ρ ranges 0.74–0.93 (mean 0.84), surviving
 84 Holm-corrected permutation testing (Figure 1). A pooled label-plus-benchmark log-cost fixed-effects model
 85 explains $R^2=82.0\%$ (adjusted 72.7%); once label and benchmark propensity are removed, **no high-overlap**
 86 **residual-cost association survives** a 20,000-draw permutation test.

87 Leave-one-benchmark-out (LOBO) prediction succeeds for CORE ($\rho=0.78$, $N=14$, $p=0.0016$), GAIA (0.87,
 88 $N=16$), SWE-mini (0.90, $N=16$), TAU (0.83, $N=14$), all robust to every leave-one-label-out omission (e.g.,
 89 CORE [0.73, 0.91], SWE-mini [0.88, 0.92]). It fails for SciCode (unstable, $[-0.50, 0.14]$; 2.1% mean accuracy,
 90 one-third zero-accuracy labels) and ScienceAgentBench (directionally negative, $[-0.77, -0.43]$; narrowest log-
 91 cost dispersion, $N=7$).

92 **Interpretation:** the portable component is a stable label-level propensity—plausibly provider pricing,
 93 reasoning-mode defaults, or routing policy—that a router could learn and reuse. But public data cannot identify

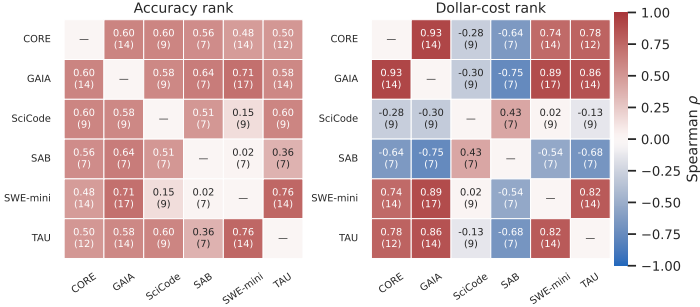


Figure 1: **Raw cost transfers; success-adjusted cost does not.** Pairwise Spearman rank correlations across benchmark pairs sharing ≥ 12 displayed labels. Raw dollar cost is strongly portable (mean 0.84); cost per expected success is not (mean 0.59), and only three CPS pairs survive Holm-corrected permutation testing.

94 which mechanism it is, because none of the distinguishing fields (token components, routing decisions, cache
95 treatment) are exposed.

96 3.3 Success-adjusted cost is a different, less portable ranking

97 Cost per expected success, $CPS_{i,b}(\epsilon) = C_{i,b} / \max(A_{i,b}, \epsilon)$ with primary floor $\epsilon=0.01$ (sensitivities at 0.001,
98 0.05, and zero-exclusion in the supplement), has mean $\rho=0.59$. Only three pairs survive correction: CORE–
99 GAIA (0.87, [0.63, 0.96]), CORE–TAU (0.83, [0.31, 1.00]), GAIA–TAU (0.78, [0.42, 0.94]). The clearest
100 raw-vs-CPS gap is GAIA–SWE-mini (0.60, [0.14, 1.13]): a label cheap in dollars is not reliably cheap per ex-
101 pected success elsewhere. No label tested on ≥ 5 workloads sits on every accuracy–cost frontier under discrete
102 nondominance, convex hull, or 5% tolerance (o4-mini Low reaches 5/6 under tolerance; no universal winner
103 under any strict definition).

104 **Implication:** routing on displayed dollar cost optimizes a signal that decorrelates from efficiency exactly where
105 procurement decisions matter.

106 4 Study 2: What Matched Trajectories Reveal That Aggregates Cannot

107 HAL’s own trace archives are encrypted, so Study 1 cannot say *why* costs differ. We use an external unen-
108 crypted corpus—977 exact task–model pairs from Open-SWE-Traces for MiniMax-M2.5 under two scaffolds
109 (OpenHands, SWE-agent), a documented four-shard boundary sample that is reproducible but not random. The
110 schema exposes resolved status, roles, tool calls, and text but not billed tokens or dollars; we treat turns, tool
111 calls, and characters as trace-burden proxies.

112 4.1 Scaffold choice changes observable burden at fixed task

113 OpenHands has lower matched medians: 123 vs. 149 trajectory turns (ratio 0.85, [0.82, 0.87]), 61 vs. 74 tool
114 calls (0.85, [0.82, 0.86]), 176,708 vs. 194,263 characters (0.95, [0.92, 0.98]); it is lower on 69.7% of pairs
115 (Figure 2a). Resolved status shows no detected difference ($p=0.197$): scaffolds differ in execution structure
116 without a detected accuracy trade-off here.

117 4.2 The “expensive failure” premium is not detected under exact matching

118 Comparing failed vs. resolved trajectories *across* tasks reproduces SWE-Eff’s pattern (19–36% heavier fail-
119 ures). But among the **107 discordant pairs**—identical task, identical model, one scaffold resolved and the other
120 failed—the failed side is heavier only 47–52 times out of 107 on every metric (two-sided binomial $p \geq 0.25$; ob-
121 served fraction CIs [0.37, 0.57] raw, [0.43, 0.63] normalized). Normalizing each trajectory by its scaffold’s
122 median burden gives failed/resolved ratios of 0.98–1.06 spanning 1. A pre-specified power calculation gives
123 80% power against a premium fraction ≥ 0.65 (failed heavier on 70+ of 107); observed 0.47–0.53 rules out
124 strong versions but not modest ones ($< 15pp$). One model, nonrandom sample: this tests the heuristic here, not
125 its general absence.

126 **Interpretation:** hard tasks make agents both burn more *and* fail more. An unconditioned failure multiplier
127 measures *which tasks were attempted*, not a property of failing runs. Task-conditioned accounting requires
128 exactly the joint schema no single public source provides.

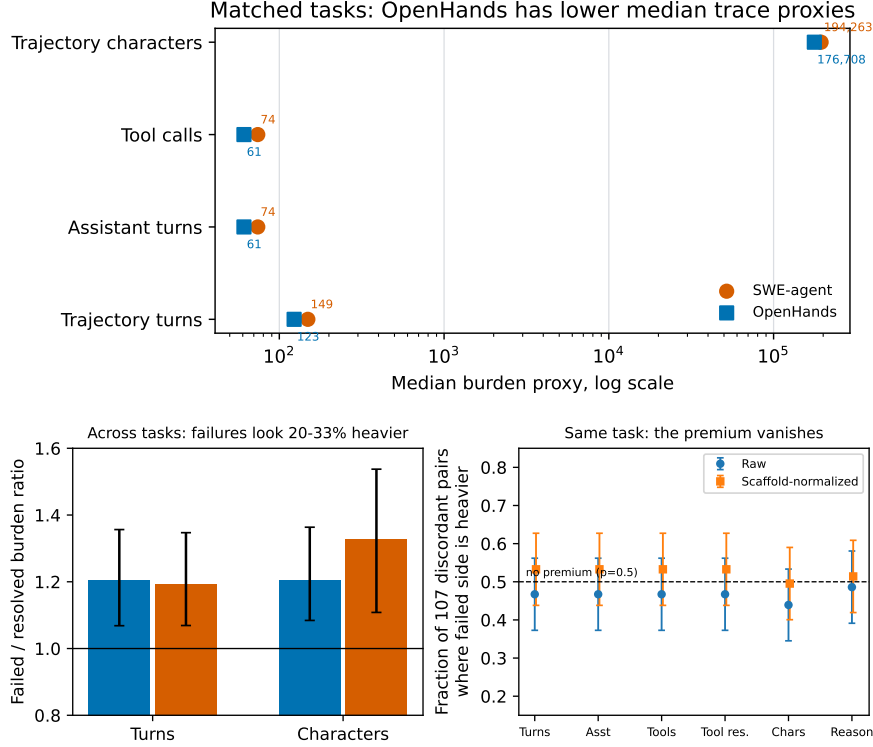


Figure 2: **(a)** Scaffold burden on 977 exact task-model pairs: OpenHands uses $\sim 15\%$ fewer median turns/tool calls than SWE-agent at fixed task and model. **(b)** The failure premium is not detected under exact task matching (80% power against premiums $\geq 15\text{pp}$). Left: naive cross-task failed/resolved ratios reproduce folklore (19–36% heavier failures). Right: among 107 discordant pairs (same task, opposite outcomes), the failed side is heavier only 47–52/107 on every metric.

Table 1: **Telemetry anatomy across two independent corpora.** Cache metrics are computed within each corpus; a share and a total-volume ratio are different quantities and not interconvertible.

	TraceLab	MIMO Claude Code
Sessions / LLM rounds	8,058 / 665,453	1,017 / 4,690
Median input / output tokens per round	132,092 / 249	n/a (constant-tuple audit)
Cache-read vs. uncached input	93.7% mean share ($\approx 14.9\times$)	753.0M / 46.3M tokens = $16.3\times$
Median final-vs-initial context	$3.6\times$ (P90 $9.9\times$)	n/a
Top-1% / top-10% input-token share	48.3% / 88.2%	— / 37.9%
Error cascade $P(\text{err} \text{err})/P(\text{err} \text{succ})$	$21.0/4.9\% = 4.29\times$	$4.3/1.0\% = 4.33\times$
Errored calls retried same-tool / retry succeeds	88.3% / 77.9%	—
Consecutive-error run length (median/P90/max)	1 / 3 / 368	—
Tool-call latency, success vs. error (median)	140 ms / 313 ms	—
Tool-result size, success vs. error (median chars)	602 / 208	—

129 5 Study 3: The Telemetry an OS Layer Would Need to See Directly

130 Study 1 shows that aggregate cost cannot explain *why* an agent is expensive; Study 2 shows that execution
 131 structure varies across agent systems at fixed task and model. This section shows that the execution variables
 132 hidden beneath aggregate cost are themselves highly heterogeneous, heavy-tailed, and state-dependent. Setting
 133 aside outcome and billing entirely: what does raw execution telemetry look like? We audit two independent
 134 coding-agent corpora (Table 1). **TraceLab**: 8,058 real-world sessions, 665,453 LLM rounds, 743,819 tool
 135 calls. **MIMO Claude Code**: 1,017 sessions, 4,690 rounds, single model. Neither links usage to outcome or
 136 billing; all summaries are compositional.

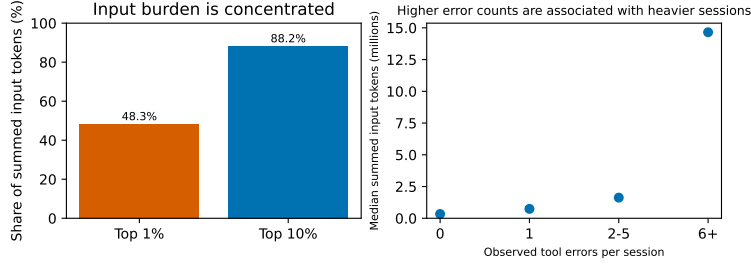


Figure 3: **Heavy-tailed burden.** Top-1%/top-10% concentration of summed input tokens (left axis) and median session input by observed-error bin (right): 0.34M at zero errors rising to 14.65M at 6+.

137 **Context is the dominant resource, not generation.** The median TraceLab round carries 132,092 input tokens
 138 against 249 output tokens (93.7% mean cache-read share); median final-round context is $3.6\times$ initial (P90
 139 $9.9\times$). A leaderboard dollar total cannot distinguish price-driven from cache-policy- or accumulation-driven
 140 spend. Tool verbosity is one mechanism: when a round carries $<1k$ tool-result characters, median next-input
 141 growth is 443 tokens; at $1k\text{--}10k$ it is 1,412; above $10k$ it is 5,420.

142 **Resource burden is heavy-tailed.** The top 1% of sessions hold 48.3% of summed input volume; top 10% hold
 143 88.2% (Figure 3). Median session input rises from 0.34M tokens at zero observed tool errors to 14.65M at six
 144 or more. Mean-based budgeting would fail to represent this concentration: resource demand is dominated by a
 145 small tail of sessions and grows sharply with observed errors (Figure 3).

146 **Error cascades replicate across independent sources.** After a successful tool call, the next call errs 4.9%;
 147 after an error, 21.0% ($4.29\times$). MIMO—different model, different site—replicates at $4.33\times$ (4.3% vs. 1.0%).
 148 Retrying is the norm (88.3% of errored calls get a same-tool retry; 77.9% succeed), but the tail is pathological:
 149 consecutive-error runs have median length 1, P90 3, max 368. Failures are also slower yet less informative:
 150 median latency 313 ms vs. 140 ms (P95 85.6 s vs. 15.8 s) while returning 208 vs. 602 characters—more wall
 151 time, less signal.

152 **Data-quality boundary.** Every released MIMO session repeats one constant per-round usage tuple, so within-
 153 session token dynamics cannot be validated there; we restrict MIMO to session counts, corpus totals, cross-
 154 session concentration, and content-derived error sequences.

155 6 An Evidence-Derived Joint Telemetry Schema

156 Table 2 summarizes the coverage gap directly. No column is complete; the missing rows are precisely those
 157 Sections 3–5 show to be load-bearing. Each field below is tied either to an analysis above or to the OS con-
 158 trol function it unblocks; *evidence-derived* means exactly that—we claim usefulness and necessity for these
 159 functions, not formal minimality. The schema per execution unit is:

- 160 1. **Task and model identity** (exact IDs, not display strings)—enables exact-matched designs; without it the
 161 Section 4 test is impossible.
- 162 2. **Scaffold/harness identity and version**—attributes burden differences to scaffold design rather than model
 163 behavior; input to routing.
- 164 3. **Configuration and policy identity** (reasoning mode, sampling parameters, token/tool budgets, routing and
 165 cache policy)—distinct from model identity; Study 1’s unexplained propensity component is plausibly the
 166 shadow of these missing fields.
- 167 4. **Outcome/resolved status**—conditions any cost statistic on success, closing the CPS portability gap.
- 168 5. **Token components per round (input, output, cache-read, cache-creation)**—separates price from accu-
 169 mulation and cache policy; lets a memory layer budget context growth.
- 170 6. **Tool-call counts, retries, error status per call**—detects and prices $4.29\times$ cascade dynamics; a design mo-
 171 tivation for governance policies that halt spiraling sessions—motivated here, not experimentally validated.
- 172 7. **Latency per call and per session**—present only as unlinked tool-call durations in one corpus; required for
 173 real-time scheduling.
- 174 8. **Billing, linked to all of the above**—the one field HAL exposes well, and exactly the field that is uninter-
 175 pretable without the others.

176 No public source we examined exposes this joint schema. We do not propose a storage format or API—a
 177 systems-design question outside our scope—but the measured consequences of its absence argue it belongs at
 178 a shared operating-system layer rather than being re-derived piecemeal inside each benchmark, where today it
 179 is not derived at all.

Table 2: **The observability gap.** Field coverage per public source we examined: ■ exposed, □ partial (display-label-only, degraded fidelity, or inferable), – absent. No examined source covers the joint schema; the union still leaves latency unlinked and round-level token fidelity unverifiable outside one corpus.

Field	HAL	Open-SWE	TraceLab	MIMO
Task identity	□	■	–	–
Model identity	□	■	□	■
Scaffold identity/version	■	■	□	■
Configuration / policy identity	□	–	–	–
Outcome / resolved status	■	■	–	–
Tokens / cache components	–	–	■	□
Retries / error status	–	□	■	□
Latency	–	–	■	–
Billing	■	–	–	–

7 Limits

This is a secondary-data analysis with clear construct boundaries. Display labels do not fully identify prompts, tools, budgets, harness versions, or routing; the HAL cohort is sparse, unbalanced, mostly single-run. Propensity adjustment is observational decomposition, not list-price control; CPS is an aggregate ratio, not the mean cost of successful runs. Study 2 uses one model and a nonrandom four-shard sample whose proxies are not tokens or dollars; the null is powered only against strong premiums (≥ 15 pp). Study 3 is observational, unlinked to outcomes or billing; MIMO’s per-round fields are unverifiable and used only where supported; design implications drawn from Study 3 (tail-aware budgeting, cascade-triggered halting) are motivations, not evaluated policies. None of this invalidates the schema argument—each limit is itself an instance of the missing-field problem the paper documents.

8 Conclusion

Across three complementary studies, we find that commonly used aggregate resource signals cannot be reliably attributed to workload, execution, or outcome without telemetry fields that the public evaluation sources we examined do not jointly expose. Raw-cost rankings transfer across workloads but largely reflect persistent displayed-label cost propensity; success-adjusted cost reorders the ranking and survives correction for half the tested pairs; the widely assumed failure-cost premium is not detected once task difficulty is held fixed by exact matching. Beneath all three results sits one missing artifact: no examined public source jointly links task, model, scaffold and configuration, outcome, token/cache usage, retries, latency, and billing. Because the layer that controls execution and allocates resources is the natural owner of exactly this telemetry, we argue the schema belongs at a shared operating-system layer for agentic AI—the role resource accounting plays in conventional OSes—rather than being reconstructed independently, incompletely, and incompatibly inside each leaderboard or trace release.

Checklist

- 1. Claims:** Supported by analyses of public data with pre-specified tests, permutation/Holm correction, and explicit power statements (Section 4).
- 2. Limitations:** Section 7 covers construct boundaries, sampling, proxy validity, and power.
- 3. Data:** All sources public: HAL CSV (Ndzomga 2026), Open-SWE-Traces (Ahmad et al. 2026), TraceLab, MIMO.
- 4. Code:** Pipeline and frozen outputs released upon publication; reproducible from public data alone.
- 5. Ethics/anonymity:** Double-blind; secondary analysis of public evaluation artifacts; no human subjects.

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